BIOL-1 Practice Test 03 — Answer Key

Exam 03 Preparation (Modules 12–16)

Part A: Multiple Choice Answers

Q	Answer
1	C
2	A
3	B
4	A
5	C
6	C
7	A
8	C
9	B
10	C
11	B
12	B
13	A
14	D
15	C
16	B
17	D
18	D
19	A

Q	Answer
20	B
21	D
22	A
23	D
24	A
25	D

Part B: Fill in the Blank Answers

Q	Answer
26	K (or carrying capacity)
27	allele frequencies
28	prezygotic
29	density-dependent
30	gene flow
31	homologous
32	decomposers
33	stabilizing selection
34	allopatric
35	sympatric

Part C: Short Answer Key

36. (Module 12) Fitness and evidence: - **Fitness:** (Accept variations.) Evolutionary **fitness** = **relative reproductive success** (passing alleles to the next generation), not just strength or long life with no offspring. - **Evidence example:** (Any one, well explained.) **Fossils** show historical sequences and transitions; **homologous** structures show shared ancestry with modified form; **biogeography** shows patterns consistent with past geography; **molecular** comparisons show more similar DNA in closer relatives.

37. (Module 13) Drift and selection: - **Bottleneck:** a catastrophic **random** die-off; survivors may have **unrepresentative** allele frequencies (not mainly "fittest for all futures"). - **Founder:** a **small** group starts a new population; the initial new pool contains **only** founder alleles, until mutation or later gene flow adds variants. - **Modes: Directional** = average shifts toward one extreme; **stabilizing** = middle favored; **disruptive** = two extremes favored, often middle disfavored.

38. (Module 14) Barriers and speciation: - **Prezygotic examples:** different mating seasons, courtship, habitat use, or gametic incompatibility before a zygote. - **Postzygotic examples:** hybrid inviability, hybrid sterility (e.g. mule), or hybrid breakdown in later generations. - **Allopatric** = geographic separation then divergence; **sympatric** = new species in **same** area, often (in plants) via **polyploidy** or strong ecological/behavioral split.

39. (Module 15) Limits and energy: - **Density-dependent:** impact **per** organism or transmission rate can rise with **crowding** (e.g. some diseases, competition for food). **Density-independent:** a similar **fraction** hit regardless of density (e.g. many fire/freeze/drought examples as framed). - **10% rule:** only ~**one-tenth** of energy stored at one level is passed to the next; many steps mean **repeated** reduction; apex consumers need a **larger** base of primary production compared with eating **producers** directly for the same food energy.

40. (Module 16) Capstone systems synthesis: - Strong answers connect at least three ideas in a chain, for example: genetic variation affects a trait; the trait changes survival/reproduction under a specific environment; population changes then alter resource use or trophic interactions. - Include a relevant feedback or constraint, such as resource competition limiting population growth, and explain its effect. - Require an explicit evidence statement such as

observed trait frequencies, survival/reproduction data, food-web observations, or population counts.

End of Answer Key